

MongoDB Practical Tasks Documentation (1–20)

This document contains the practical implementation of 20 MongoDB tasks performed using MongoDB Community Server and Mongosh on Windows. The project covers CRUD operations, indexing, aggregation, validation, replica sets, sharding, backup, cluster checking, and security commands.

Task 1 – MongoDB Installation

Objective: Installed MongoDB Community Server and Mongosh on Windows.

Commands Used:

```
mongod --version
mongosh --version
```

Explanation: Verified MongoDB server and shell installation successfully.

Task 2 – Create Database

Objective: Created and switched to a new database.

Commands Used:

```
use studentDB
```

Explanation: MongoDB creates the database when data is inserted.

Task 3 – Create Collections

Objective: Created collections for storing documents.

Commands Used:

```
db.createCollection("students")
db.createCollection("courses")
```

Explanation: Collections are similar to tables in relational databases.

Task 4 – Insert Data

Objective: Inserted student and course records.

Commands Used:

```
db.students.insertOne({...})
db.students.insertMany([...])
```

Explanation: Used insertOne and insertMany to add documents.

Task 5 – Update Data

Objective: Updated student documents.

Commands Used:

```
db.students.updateOne(...)  
db.students.updateMany(...)
```

Explanation: Modified single and multiple documents.

Task 6 – Delete Data

Objective: Deleted records from collection.

Commands Used:

```
db.students.deleteOne(...)  
db.students.deleteMany(...)
```

Explanation: Removed matching documents from collection.

Task 7 – Create Index

Objective: Created index on student name field.

Commands Used:

```
db.students.createIndex({ name: 1 })
```

Explanation: Indexing improves query performance.

Task 8 – Query Documents

Objective: Queried documents using conditions.

Commands Used:

```
db.students.find({ name: "Aarav" })
```

Explanation: Retrieved filtered data using find().

Task 9 – Cursor

Objective: MongoDB returned cursor objects for queries.

Commands Used:

```
db.students.find().pretty()
```

Explanation: Cursor helps navigate query results.

Task 10 – Aggregation

Objective: Grouped and analyzed records.

Commands Used:

```
db.students.aggregate([...])
```

Explanation: Aggregation pipeline processes data for analysis.

Task 11 – Lookup (Join)

Objective: Joined students and courses collections.

Commands Used:

```
$lookup
```

Explanation: Used lookup to simulate relational joins.

Task 12 – Filter Data

Objective: Filtered records using operators.

Commands Used:

```
db.students.find({ age: { $gt: 20 } })
```

Explanation: Used operators like \$gt for filtering.

Task 13 – Sort Data

Objective: Sorted records ascending and descending.

Commands Used:

```
db.students.find().sort({ age: 1 })  
db.students.find().sort({ age: -1 })
```

Explanation: Sorting arranges documents in required order.

Task 14 – Validation

Objective: Created validated collection.

Commands Used:

```
db.createCollection("validatedStudents", {...})
```

Explanation: Validation ensures proper document structure.

Task 15 – Replica Set

Objective: Configured replica set using separate terminals.

Commands Used:

```
Terminal 1:
mongod --replSet rs0 --port 27018 --dbpath C:\data\db2
Terminal 2:
mongosh --port 27018
rs.initiate()
rs.status()
```

Explanation: Replica set provides redundancy and high availability by maintaining copies of data.

Task 16 – Sharding

Objective: Configured sharding architecture using four terminals.

Commands Used:

```
Terminal 1:
mongod --configsvr --replSet configReplSet --port 27019 --dbpath C:\configdb
Terminal 2:
mongosh --port 27019
rs.initiate()
Terminal 3:
mongos --configdb configReplSet/localhost:27019 --port 27020
Terminal 4:
mongosh --port 27020
sh.addShard("rs0/localhost:27018")
sh.status()
```

Explanation: Sharding distributes data across servers for scalability.

Task 17 – Backup

Objective: Created MongoDB backup using database tools.

Commands Used:

```
.\mongodump.exe --db studentDB --out C:\Users\HP\backup
```

Explanation: Backup stores BSON copies of collections.

Task 18 – Automated Backup

Objective: Created backup batch command for automation.

Commands Used:

```
mongodump.exe --db studentDB --out C:\Users\HP\dailybackup
```

Explanation: This command can be scheduled using Windows Task Scheduler.

Task 19 – Cluster Status

Objective: Checked replica cluster status.

Commands Used:

```
rs.status()
```

Explanation: Verified health and status of replica set.

Task 20 – Security / Configuration

Objective: Checked MongoDB server configuration.

Commands Used:

```
db.adminCommand({ getCmdLineOpts: 1 })
```

Explanation: Displays server configuration, paths, ports, and startup options.

Conclusion:

This project demonstrated practical MongoDB operations including CRUD, indexing, aggregation, validation, replica sets, sharding, backup, and configuration management. The project provided hands-on understanding of MongoDB architecture and administration concepts.